

The Missing Benchmark

Grading Prediction-Market Inflation Forecasts, and Why the Obvious Way Is Wrong

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Keywords: prediction markets, inflation forecasting, calibration, data vintages, real-time data.

JEL classification: E37 (forecasting and simulation of prices and business fluctuations);

C53 (forecasting and prediction methods); G14 (information and market efficiency); C82

(methodology for collecting and estimating macroeconomic data).

Abstract

A companion paper scores Kalshi's temperature ladders against the National Weather Service's own probabilistic guidance and finds the market better. The obvious next step is to do the same to the federal statistical agencies. It cannot be done, and the reason is worth stating: the Bureau of Labor Statistics produces the truth and publishes no forecast of it. No federal body publishes a probability distribution over any quantity Kalshi settles monthly. The benchmark that made the weather comparison possible does not exist here.

This paper reports what can be done instead, and one thing that must be done before anyone tries.

Against the Cleveland Fed's inflation nowcast, the only free, vintaged, official-sector forecast of the exact settled quantity, Kalshi's CPI ladders are the better forecast. On 1,047 contracts across 53 monthly releases, Brier scores are 0.089 for the market one day before release against 0.137 for the nowcast, a difference of 0.049 with a release-clustered t of 3.45. The nowcast is a point forecast, so a dispersion assumption is required; the conclusion holds across every setting from 0.6 to 2.0 times the benchmark's own realised error, with t between 3.3 and 5.3. The advantage is concentrated in headline inflation, t of 3.3 on the monthly print and 4.4 on the year-over-year, and is absent in core year-over-year, t of 0.55.

The measurement warning is the more useful result and does not depend on any modelling choice. These contracts settle on the **first print**. BLS re-estimates seasonal factors every February and revises five prior years, so the seasonally adjusted series available today is not the series the contracts were graded against. Of 155 months in the archive, **89, or 57 per cent, fall in a different one-tenth-of-a-point bin under the current vintage than under the first print**. December 2021 first printed at +0.470 and reads +0.691 today: a contract on "above 0.5 per cent" settled no and would be scored yes. Any study grading these contracts from a current data pull is silently mis-scoring more than half its sample.

1. Introduction

The companion paper to this one asks whether a prediction market forecasts better than the agency whose job it is to forecast, and answers it for daily maximum temperature, where the National Weather Service publishes an operational probabilistic forecast of exactly the quantity the contracts settle on.

Macroeconomic releases look like the natural extension. Kalshi lists threshold ladders on the monthly CPI print, on core CPI, and on both year-over-year rates. The releases are scheduled, the settlement source is a named federal statistic, and the subject is one economists care about. The extension does not work, and the reason is instructive rather than merely inconvenient.

The BLS does not forecast. It measures. It publishes a point estimate of what already happened, on a schedule, and never a probability distribution over what it is about to publish. The same is true of every other federal statistical agency. So the question the weather paper asks, whether the licensed producer of authoritative probability is any good at it, has no defendant here: there is no official probability to score.

What exists instead is a research nowcast. The Cleveland Fed publishes a daily-updated point estimate of the current month's CPI and core CPI, built from a deliberately simple bridge model. It is not the BLS's forecast, it is not anyone's official forecast, and beating it is a lower bar than beating the National Blend of Models. Section 3 scores against it anyway, because it is the only free, vintaged, official-sector forecast of the exact settled quantity, and Section 6 is explicit about how much weight the result can carry.

The contribution this paper is confident about is a measurement warning. Grading a settled contract requires knowing what it settled on. For CPI that is the first print, and the first print is not what a data pull returns today. Section 5 shows that more than half of the months in the archive would be assigned to a different bin by the naive procedure, and identifies the one public source that solves it.

The contract structure differs from the weather paper's and is worth stating up front. Kalshi's CPI ladders are threshold ladders, not bucket ladders: each leg pays if the print exceeds a strike, so the legs are nested survival probabilities that decline monotonically and do not sum to one. The benchmark for a leg is therefore a survival probability under the nowcast's error distribution, and the legs of one release are mechanically linked to a single realised number, so all inference clusters on the release.

2. Data

Markets. Every resolved Kalshi contract on headline CPI, core CPI, and their year-over-year counterparts, requiring at least three trades: **1,047 contracts across 164 events and 53 monthly releases**, for target months from October 2021 to September 2025.

Series	contracts	releases
CPI, monthly	334	53
CPI, year-over-year	287	36
Core CPI, monthly	231	41
Core CPI, year-over-year	195	34

The benchmark. The Cleveland Fed's inflation nowcast, taken as the last value published strictly before the release. The public nowcasting page is backed by a chart endpoint carrying, for each of 158 target months back to July 2013, the full daily sequence of nowcasts made for that month together with the release-day actual. That daily sequence is what makes the comparison possible: it gives the forecast as of the date the market traded rather than a single current value.

The archive is a genuinely frozen real-time record rather than a backfilled reconstruction. Two Wayback snapshots of the same endpoint, from November 2022 and March 2024, were diffed against the live file: 6,614 and 7,522 nowcast values respectively, none of them changed.

Turning a point into a distribution, and the assumption that requires. The nowcast has no published uncertainty: no standard deviation, no percentiles, no fan chart. To score it as a probability forecast a dispersion must be attached, and this paper attaches the benchmark's own realised error, the root mean squared difference between its final pre-release nowcast and the first print. That is 0.187 percentage points for the monthly headline

print, 0.156 for core, 0.203 for headline year-over-year and 0.170 for core year-over-year. A leg's benchmark probability is then the survival probability of a normal centred on the nowcast with that dispersion.

This is the paper's one free parameter and Section 4 reports its influence rather than defending the choice.

Timing. Nowcasts post each business day at about 10:00 Eastern; the contracts trade until the release. The market therefore strictly nests the nowcast's information set, exactly as in the weather paper, which is why Section 4's encompassing regression rather than the raw score is the defensible claim.

3. The market is the better forecast

Table 1. Brier scores, by how far before the release the price is taken.

Forecast	Brier	contracts
Kalshi, last trade	0.08563	1,047
Kalshi, one day before release	0.08930	1,038
Kalshi, three days before	0.09233	1,030
Kalshi, seven days before	0.09720	1,010
Cleveland Fed nowcast	0.13733	1,047

Table 2. Head-to-head, standard errors clustered on the release.

Market price taken	difference	s.e.	t	95% interval	releases
Last trade	0.05170	0.01372	3.8	[0.0278, 0.0784]	53
One day before	0.04909	0.01423	3.5	[0.0236, 0.0770]	53
Three days before	0.04623	0.01445	3.2	[0.0205, 0.0749]	53
Seven days before	0.03896	0.01384	2.8	[0.0144, 0.0673]	53

The market is ahead at every horizon and the gap widens as the release approaches, which is what a market absorbing information should do. The one-day price is used as the headline because it is the last point at which the comparison is clean.

Table 3. By series. The advantage is not uniform, and where it is absent is informative.

Series	market	benchmark	difference	t	releases
CPI, year-over-year	0.05868	0.12054	0.06276	4.4	36
CPI, monthly	0.09036	0.15977	0.06942	3.3	53
Core CPI, monthly	0.10173	0.13510	0.03506	2.0	41
Core CPI, year-over-year	0.11801	0.12622	0.01016	0.6	34

The market's edge is on **headline** inflation and disappears on core. That ordering is the opposite of what a story about general market intelligence would predict, and it fits a specific one: headline inflation carries the volatile

energy and food components, which move with observable daily prices that a trader can watch and a monthly bridge model cannot. Core is the smoother, more forecastable series, and there the nowcast very nearly matches the market.

4. The one free parameter, and whether the market relays the nowcast

Table 4. Sensitivity to the assumed dispersion. The benchmark is rebuilt with its error standard deviation scaled, and rescored.

dispersion	benchmark Brier	difference	t
0.6×	0.14579	0.05649	3.4
0.8×	0.14048	0.05118	3.3
1.0×	0.13839	0.04909	3.5
1.2×	0.13839	0.04908	3.7
1.5×	0.14069	0.05138	4.2
2.0×	0.14727	0.05797	5.3

The benchmark scores best at essentially the dispersion its own errors imply, and the conclusion survives every setting tested: the difference never falls below 0.039 and the t never below 3.3. The free parameter is not doing the work. Had the result depended on it, this paper would not be worth writing.

Table 5. Forecast encompassing. Outcome on both forecasts in log-odds, clustered on the release.

	coefficient	s.e.	t
Market, one day before	0.9088	0.0999	9.1
Cleveland Fed nowcast	0.3058	0.1133	2.7

The market survives controlling for the nowcast and the nowcast survives controlling for the market, with the market carrying roughly three times the weight. This is a weaker encompassing result than the weather paper's, where the benchmark's coefficient collapses to 0.16, and the honest reading is that the nowcast retains genuine independent information about inflation in a way NBM does not about tomorrow's temperature.

5. The measurement warning

The results above depend on modelling choices. This section does not.

These contracts settle on the first print. A Kalshi CPI contract resolves against the number the BLS publishes on release day. The BLS then re-estimates seasonal factors every February and revises five prior years of the seasonally adjusted series. The consequence is that the series a researcher pulls today is not the series the contracts were graded against.

Table 6. First print against the current vintage, monthly headline CPI.

	value
Months compared	155
Months assigned a different 0.1pp bin	89

Share	57.4%
Mean absolute revision	0.0613 pp
Largest absolute revision	0.2413 pp

Examples make the size concrete. December 2021 first printed at +0.470, which settles a contract on "above 0.5 per cent" as **no**; today the same month reads +0.691, which would score it **yes**. March 2017 first printed at -0.288 and now reads -0.047. March 2016 first printed at +0.090 and now reads +0.313.

More than half the archive is affected, the revisions are large relative to the 0.1 percentage point strike spacing these ladders use, and the direction is not random, since seasonal re-estimation redistributes within a year. A study grading these contracts from a current data pull is mis-scoring the majority of its sample, and no amount of care in the econometrics recovers from that.

The fix is available and free. Because the Cleveland Fed archive is frozen, its "Actual" series is the first print. One file therefore supplies both the benchmark and the correct settlement vintage. Every result in this paper uses it. Its first prints were spot-checked against the BLS news releases published on the day.

6. What this does and does not establish

It does not indict the federal statistical agencies. The BLS publishes no forecast, so it is not being scored. The Cleveland Fed nowcast is a research product from a different institution running a deliberately simple model, and it is not the consensus: release-eve survey consensus typically carries roughly half the nowcast's error. Beating the nowcast is a lower bar than beating the professional forecasting community, and this paper does not claim to have cleared the higher one.

It rests on an assumed dispersion. The benchmark is a point forecast. Table 4 shows the conclusion is stable across the range, but a genuine distributional benchmark would be better and none exists.

The sample is small. Fifty-three releases, against the companion paper's 6,023 city-days. The releases are closer to independent than consecutive weather days are, which helps, but the confidence intervals are correspondingly wide and the core year-over-year result is a null rather than a finding.

It supports no trading claim. Scores are Brier, not returns.

What it does establish is that where an official probabilistic benchmark exists the comparison is worth making, that where one does not the best available substitute still favours the market on headline inflation, and that the standard way of grading these contracts is wrong for the majority of months.

7. Conclusion

There is no macro analogue of the National Weather Service. No federal statistical agency publishes a probability distribution over the number it is about to release, so the question the companion paper asks cannot be asked of them.

Against the only free, vintaged, official-sector nowcast of the same quantity, Kalshi's CPI ladders score better, by 0.049 in Brier with a release-clustered t of 3.45, robust to the dispersion assumption, concentrated in headline inflation and absent in core.

The finding this paper would defend hardest is the smallest one. Contracts on revised statistics must be graded on the vintage they settled against, and for the CPI more than half of all months change bin between the first

print and the series a researcher would otherwise use.

Data, provenance and replication

Market data are the public Kalshi market and trade records from a local parquet snapshot taken 23 November 2025. Benchmark and settlement data are the Cleveland Fed inflation nowcasting archive, retrieved from the chart endpoint behind the Bank's public nowcasting page. The current-vintage comparison of Section 5 uses CPIAUCSL from FRED.

Every estimate is produced by deterministic Python in two modules, ``pipeline/fetch_cleveland.py`` and ``pipeline/cpi_lab.py``, whose output is written to a machine-readable record. A check walks this manuscript, extracts every number, and fails if any of them is absent from that record. Language models were used to draft prose, to search the literature and to attack the argument; they computed nothing.

Disclosure statement. The author is an independent researcher. This work received no external funding. The author has never held an account or position on Kalshi or on any other prediction-market venue, and has no financial interest in any exchange, contract or firm discussed in this paper. All data are public.

References

- Brier, G. W. (1950). Verification of Forecasts Expressed in Terms of Probability. *Monthly Weather Review*, 78(1), 1-3.
- Croushore, D., and Stark, T. (2001). A Real-Time Data Set for Macroeconomists. *Journal of Econometrics*, 105(1), 111-130.
- Diebold, F. X., and Mariano, R. S. (1995). Comparing Predictive Accuracy. *Journal of Business and Economic Statistics*, 13(3), 253-263.
- Gneiting, T., and Raftery, A. E. (2007). Strictly Proper Scoring Rules, Prediction, and Estimation. *Journal of the American Statistical Association*, 102(477), 359-378.
- Knotek, E. S., and Zaman, S. (2017). Nowcasting US Headline and Core Inflation. *Journal of Money, Credit and Banking*, 49(5), 931-968.
- Le, N. A. (2026). Decomposing Crowd Wisdom: Domain-Specific Calibration Dynamics in Prediction Markets. arXiv:2602.19520.
- Murphy, A. H. (1973). A New Vector Partition of the Probability Score. *Journal of Applied Meteorology*, 12(4), 595-600.
- Wolfers, J., and Zitzewitz, E. (2004). Prediction Markets. *Journal of Economic Perspectives*, 18(2), 107-126.